

Excel Basics

Note

I'm using is the online web version of Excel, so it may look different from the desktop version. However, the core functionalities are the same.

What is Excel?

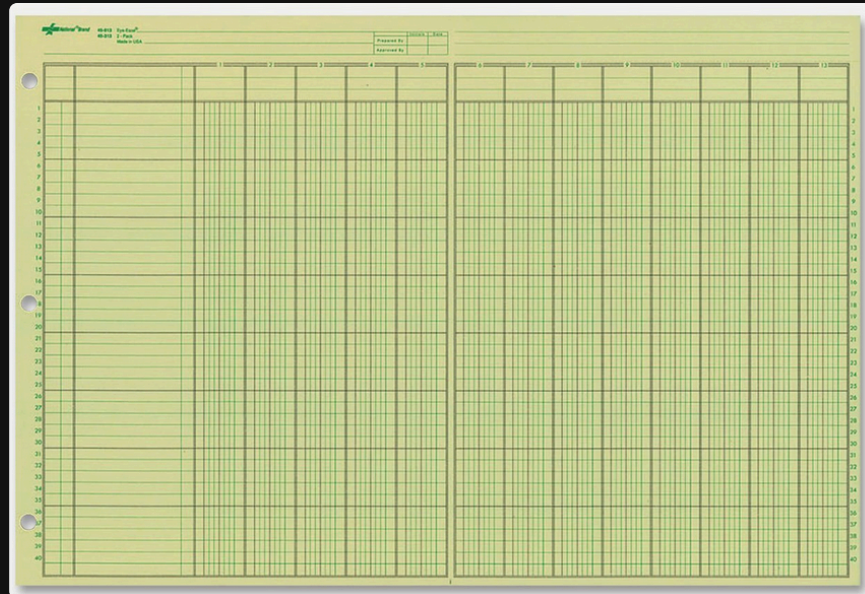
Specifically

What analog tool does Excel replace?

The analog spreadsheet

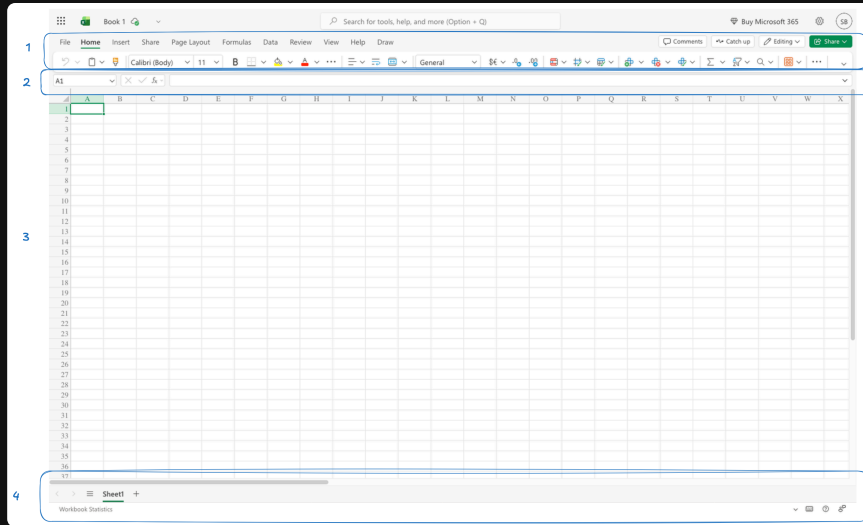
Excel is so widespread, so common, and so old that many people think of it as a *generic* term for spreadsheets, just like how we use "*Google*" to refer to any search engine, or "*Xerox*" to refer to any photocopier.

Spreadsheets are a type of document that *organizes data in rows and columns*, and allows you to perform calculations and analysis on that data.



The interface

Interface



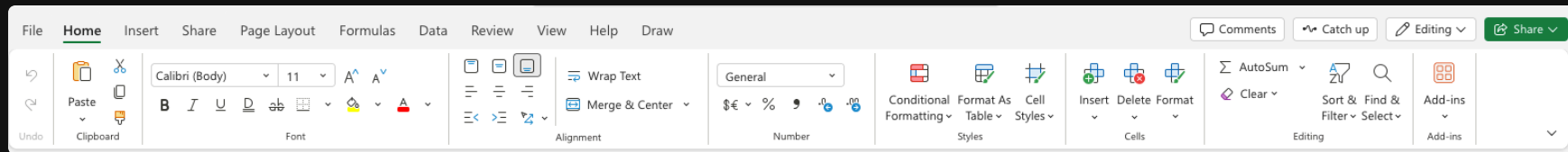
1. **Ribbon**, where you can find the *tools and features* of Excel, organized into tabs and groups.
2. **Formula bar**, where you can *enter and edit formulas* and data in the selected cell.
3. **Worksheet area**, where you can *see and interact* with your data in a grid of cells.
4. **Sheet tabs**, where you can *switch* between different worksheets in the same workbook.

Ribbon

The ribbon is a way of organizing the tools into *different tabs*, they let you ask the question

I want to change how my worksheet looks when it gets printed, where do I go?

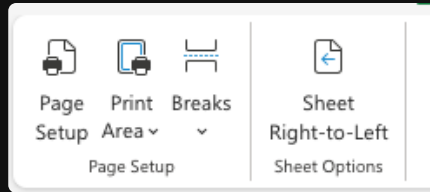
And come to a *logical guess* as to where it is, *even if you have never used Excel before*.



Of these, which one do you think contains lets me change how my worksheet looks when it gets printed?

Ribbon Groups

Each tab is further organized into *groups*, which are collections of *related tools* that are often used together.



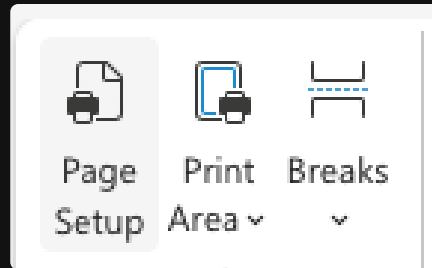
Which of these groups do you think lets me change how my worksheet *looks* when it gets printed?

Tools and Features of Excel

Excel has *thousands* of tools and features, the ribbon let's us find them easily

Let's say, I want to

1. Make sure that my worksheet is printed in *landscape* orientation, and
2. Only print the *first 10 rows* of my worksheet



Which of these should I use?

Exercise

I want to turn data into a bar chart so it's easier to read in my presentation

1. Of the main tabs, which one do you think contains the tools to create a *bar chart*?

hint: I want to *add* a bar chart to my worksheet

2. On that tabs, which group do you think contains the tools to create a bar chart?

3. Which tool do you think I should use to create a bar chart?

Exercise

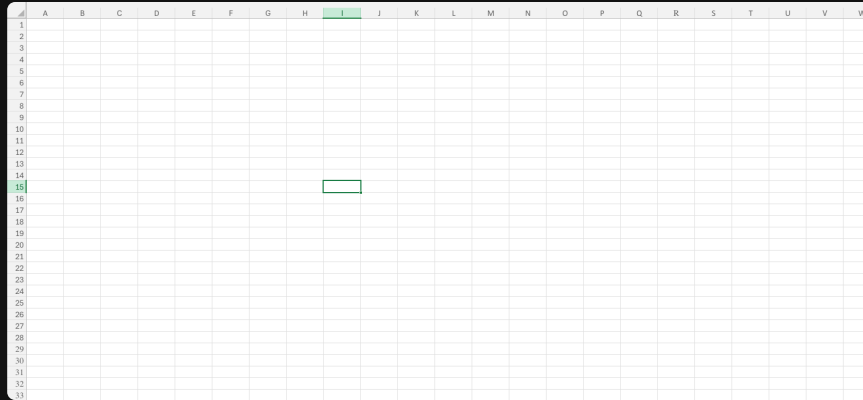
I want to alphabetize a list from a to z, and I need to remove any duplicates

1. Of the main tabs, which one seems dedicated to *managing, sorting, and cleaning* data?
2. Which ribbon group seems to be dedicated to *sorting* data?
3. Which ribbon group seems to be dedicated to *removing duplicates* from data?

Exercise

I want to make the text in this row bold, change the background color to yellow, and center the text

Worksheet Area



The worksheet area is where you can see and interact with your data in a *grid of cells*.

It's labeled with

- *letters for columns*, and
- *numbers for rows*.

Side note on importing data

Excel is able to import data from almost *any* source, such as *databases, web pages, text files*, etc.

1. Create a new blank *workbook* (an excel file)
2. What tab do you think contains the tools to import data from a text file?

On your NEO, there is a `data.csv` file, which contains some data, import it to Excel

It should look something like this:

	A	B	C	D	E	
1	Item ID	Category	Product Name	Color	Location	
2	INV-001	Electronics	Wireless Mouse	Matte Black	Desk A	
3	INV-002	Stationery	Fountain Pen	Cobalt Blue	Drawer 2	
4	INV-003	Furniture	Ergonomic Chair	Space Grey	Cubicle 4	
5	INV-004	Plant	Monstera Deliciosa	Vibrant Green	Window Sill	
6						

Exercise

1. what is the exact *cell address* of "*Ergonomic Chair*"?

Note that it's always *column letter* followed by *row number*

Exercise

2. What is in "`C2`"?

Exercise

3. You want to make all the headers in the *first row* bold (Item ID, Category, etc), what *range* would you select?

A range is a *block* of cells, and it's denoted by the *top-left cell* and the *bottom-right cell*, separated by a colon.

`A1:C1` refers to the range of cells from A1 to C1, which includes A1, B1, and C1.

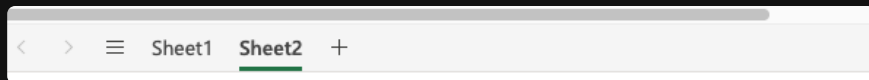
Exercise

4. You want to copy the *product names and colors* for the *first three items*, and paste them into a new worksheet
 - what is the range for that *3x2* block of data?

Exercise

5. What does *B2:B5* refer to?

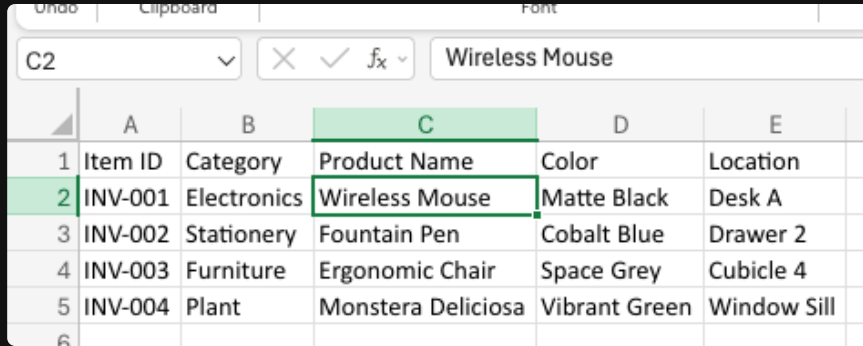
Sheet Tabs



A work*book* can have multiple work*sheets*, and you can switch between them using the sheet tabs at the bottom of the worksheet area.

This is useful for organizing your data into *different sections*, or for keeping *related data together* in the same file.

Formula Bar



The screenshot shows a spreadsheet application with a formula bar at the top. The formula bar has a dropdown menu showing 'C2', a close button (X), a checkmark, and a function button (fx). The text 'Wireless Mouse' is entered in the formula bar. Below the formula bar is a table with the following data:

	A	B	C	D	E
1	Item ID	Category	Product Name	Color	Location
2	INV-001	Electronics	Wireless Mouse	Matte Black	Desk A
3	INV-002	Stationery	Fountain Pen	Cobalt Blue	Drawer 2
4	INV-003	Furniture	Ergonomic Chair	Space Grey	Cubicle 4
5	INV-004	Plant	Monstera Deliciosa	Vibrant Green	Window Sill
6					

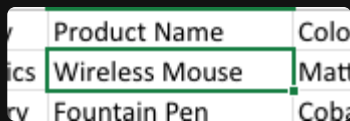
The formula bar is where you can enter and edit formulas and data in the selected cell.

You can use the formula bar to enter *any* data into a cell, such as text, numbers, dates, etc.

It's also where the data of the *currently selected cell* is displayed

Cells

What are cells?



	Product Name	Color	Category
1	Wireless Mouse	Matte Black	Electronics
2	Fountain Pen	Cobalt Blue	Stationery

Cells are "*boxes*" of data

Fundamentally, they are *containers* for *expressions* that evaluate to a *value*.

What is the difference between

1. The arabic numeral `5`
2. five objects, like `five donuts`
3. and `3 + 2`

This *model of thinking* lets us more easily understand how Excel works, and how to use it effectively.

Especially when we get started with *functions*

Assignment

The goal of this *assignment* is to get familiar with the *interface of Excel*

Create an excel file, and create a schedule

It must have the day as the headers, the time as rows, and the activities in the cells

Name the file `lastname_firstname_schedule.xlsx` i.e. (`alaan_neil_schedule.xlsx`), and submit it to NEO

- 5 pts for having a schedule with days as headers, time as rows, and activities in the cells
- 4 pts for usage of formatting tools, such as bold, background color, borders, etc
- 1 pts for correct file name